1. Exercise 2

Let A be a ring and let $A[x]$ be the ring of polynomials in an indeterminate x, with coefficients in A. Let $f = a_0 + a_1 x + \cdots + a_n x^n \in A[x]$.

Question 2.i

Prove that f is a unit in $A[x] \Leftrightarrow a_0$ is an unit in A and $a_1, \ldots, a_n$ are nilpotent.

Hint: If $b_0 + \cdots b_m x^m$ is the inverse of f, prove by induction on r, that $a_n^{r+1} b_{m-r} = 0$. Hence show that a_n is nilpotent and then use Ex. 1.

Solution Suppose we use the hint, then we have to prove that dfd